Meeting on IDP. Nov 22, '58

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Bloomer
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Personal opinions;
Pete Spratt was manager of IDP at start.
Historical. G. Hawkins committee formed March '58.
to look at IDP (1) what is needed for company to get into
(2) " do we have on shelf to get going.

Can't answer (1) precisely. need to know mkt wise.
don't know what IDP really is.
- make some guesses: studied SABRE, AMC, Boeing, SAC, Military

big diff between military & commercial — willingness to spend money,
IDP will have to come by evolution — hard to convert over
from batch to IDP
— economic advantages are not well known.
— Pete Spratt took over later Data Communications
headed with G. Hawkins

Most studied communication systems — mod-demod equipment
eg. AT&T digital
IBM & AT&T both wanted to control.

what functions of computer are needed? Stibitz,

Nutter - comm. Terminal - multiplexing area (data gathering)

Kaeli - computer area

Wong - new IDP computer design

Kaeli Studied:

305, proposed 310 (as originally proposed), 650 Ramac, 705, 7070
by San Jose X10

STRETCH. (not the 7090
(because not a commercial processor - not checked)
etc.

- in contrast to SABRE - -

Transac

American Airlines 7090 7090

concentrated on in-line processing - a multitude of several transactions

"Anaw-Hart study" approach

- multiprogramming, cueing, interrupt, priority control, relative addressing.

results: 305, 650R, 310 proposed San Jose could handle on limited ant.

because 1. internal speed, 2. size of working memory, 3. single data channel (one direction at a time.) - hard to achieve integrated processing, do "in-line" however,

305
had inquiry facility. (change & read out put in data)

705, 7070, STRETCH

control problems encountered 1. Cueing.
2. multiprograming,
3. relative addressing.
4. interrupt.

cueing: - as freq. of Transtion increases - must pile up requests -

(1) needs to smooth average no. at computer.

- at input area - multiplex references to file records,

(2) multiple file refs

- one transaction may call in a hundred records
- rather than design for worst case — share an area between programs; not a+b+c but a+ε common

(3) working areas & use cue,

- if one has integrated a large no. of programs,

one no. of progs. = 24 per customer. (daily, weekly, monthly)

one no. of math steps ≈ ~~1000~~ 1750 instrs (not data)

(Arrow-Hunt: 60 M characters, 100 diff. programs.
of working storage.)

12 M characters for Meteorology Prof Airforce

(4) ~~(5)~~ so cueing of programs is necessary also.

~~Bottleneck~~ Imbalance between I/O & computing

Multi-programming is one ~~(5)~~ solution.

facilitating multiprog (1) interrupt procedure — under control of option of prog.

store status: sto IC, status of ~~(5)~~ Triggers, accumulator,

(Honda on 205 — 15 accumulators.)

(2) record control wd. — scatter read/write — reset address,

(3) memory protection.

as of June

cost: terminal equipment is most expensive

SABRE cost of Terminals 2X cost of system. rent to

serializing — mod, demod.

keyboards: Typewriter, small keyboard 1,2,3,4, date, etc.

flite card readers.

message-by-message transmission? buffers

(4) Relative addressing. – to avoid

(5) interrupt control – automatic – (preferred system)

or – trigger to be tested

~~(6) priority co~~

Priority Control:

– some progs. are urgent – some can be deferred

– transaction

– load fluctuation

– cyclic – predictable. (I am going to cash checks)

– unpredictable.

Probs: (1) Simultaneous Record Reference (conflict)

eg. 2 channels to same Disk

– one could deny other references to same record during a read-write cycle. – by programming. – may be one protection.

(1) only one channel per file

(2) keep waiting list

(2) Dead-lock Situation:

Example: read A | B | can't use B

read B | A | can't use A

} neither can proceed.

complicated program to unravel,

must restore A to original state &

conclusion:

- a large scale computer is probably not best way — (because of degree of complexity of whole job)
- a family of 310's is probably better, (3)
  - duplication - reliability
  - simpler programming

F-60 good idea -

Terminal end - look like a tape unit - accepting DSO or Exchange.

- 705 BCD code preferred.

Memory: 305 215 drum
10K core
705 80K
9070 99900words

Ref:
W. Wittenberg, Rohmeri & Amer. Airlines

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need to "stream-line" business -- Tradition

eg. paying by cash - to nearest $1,

Boeing Job -- data gathering,
3 phases (1) card to card
(2) card to tape
(3) card to computer
200 readers.

- fitting computer to present mode of operation
- must convert to "computer philosophy"

ref: IDP
Bill Morgan

Philco: NADT ~ 6^{00} ea.

(4 stages 17 mpsec,
but with loops 11 mpsec.)